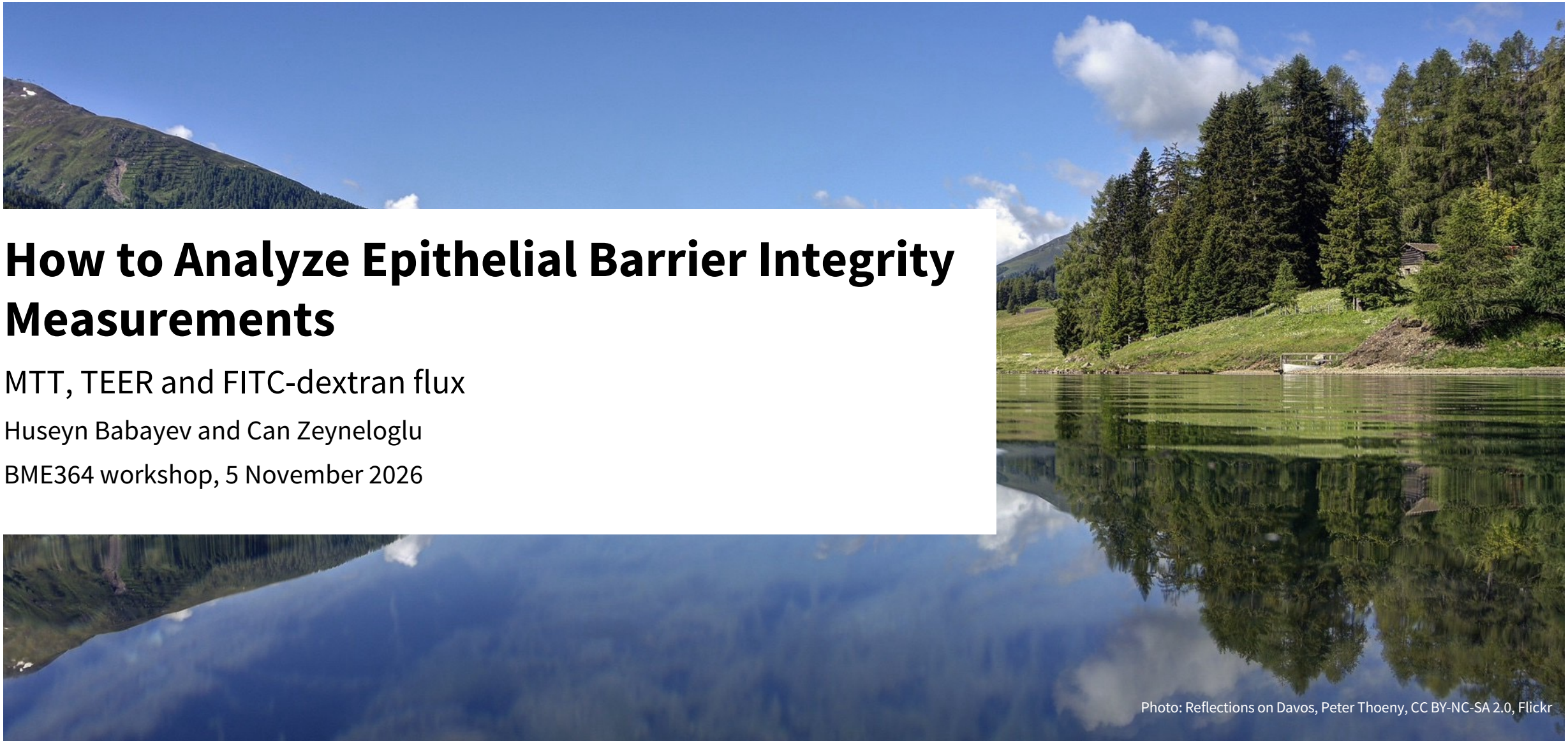
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How to Analyze Epithelial Barrier Integrity Measurements

MTT, TEER and FITC-dextran flux

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BME364 workshop, 5 November 2026

Photo: Reflections on Davos, Peter Thoeny, CC BY-NC-SA 2.0, Flickr

The Group I experiment in Davos

0 h

- TEER baseline (24-well Transwell)
- P80 apical: 0.02, 0.1, 0.5 %
- Control: medium only

24 h

- TEER
- FITC-dextran 4 kDa apical (0.5 mg/mL)

48 h

- TEER
- FD4 sampled apical + basolateral
- MTT (96-well): P80 1 → 0.03125 %

Four topics, one repository

01
Statistics
`01_statistics.Rmd`

02
MTT viability
`02_mtt.Rmd`

03
TEER
`03_teer.Rmd`

04
FITC-dextran flux
`04_flux.Rmd`

One experiment: n = inserts (TEER, flux; 4 per condition) or wells (MTT; 6 per concentration, 12 controls). Four groups → ANOVA + Dunnett vs control.

Parametric and nonparametric statistics

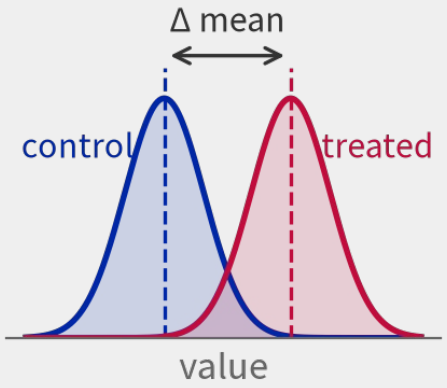
Theory, data structure, algorithm, implementation

01

Two families of tests, one logic

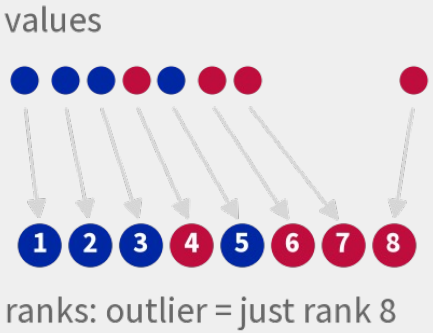
Parametric

- Normal residuals, equal variances, independence
- Compare **means**: t-test, ANOVA, mixed models
- More power when assumptions hold
- Effect: mean difference, 95 % CI



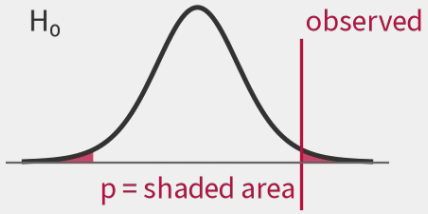
Nonparametric

- Values → **ranks**; no normality needed
- Compare **distributions**: Mann–Whitney, Kruskal–Wallis, Wilcoxon
- Robust to skew and outliers; less power at small n
- Effect: η^2_H, Hodges–Lehmann shift



Shared logic

H_0 → test statistic → p = P(data at least this extreme | H_0). α = 0.05 limits type I errors; power = 1 − β limits type II.



n in this course: technical replicates

One Caco-2 experiment → n = wells or inserts per condition. Conclusions hold for this experiment; reproducibility needs independent repeats (passages, days).

Structure decides the test

Wide format: each group has its own column

ctrl	P80
1.095	3.187
0.967	2.231

Long format: one row per well, use this one

well	group	value
W01	ctrl	1.095
W02	ctrl	0.967
W03	ctrl	1.492
W07	P80	3.187
W08	P80	2.231
W09	P80	2.045

R's tests and plots need the values in one column and the group in another, as in t_test(value ~ group). In the wide table, two values in one row look paired, but they come from different wells.

What counts as n

In one Caco-2 experiment, n is the number of wells or inserts per condition. Images and cells inside one well are not n: average them per well first, or use a mixed model.

Independent or paired groups

Each condition has its own wells, so the groups are independent. Data are paired only when the same well or insert is measured twice, for example before and after treatment, or over time.

How many groups and factors

Two groups need a t-test; three or more need an ANOVA. A second factor, such as time, needs a two-way ANOVA or a mixed model.

Decision algorithm: from design to test

1. **Unit and n**: the wells or inserts that were treated
2. **Pairing**: same well or insert measured again, or not
3. **Groups and factors**: 2 vs ≥ 3 groups; one or two factors
4. **Assumptions**: QQ plot / Shapiro–Wilk of residuals; Levene
5. **Test and post hoc**: from the table; correct for multiple tests
6. **Report**: effect with 95 % CI, exact p, n, test name

Design	Parametric	Nonparametric
2 independent	Welch t-test	Mann–Whitney U
2 paired	Paired t-test	Wilcoxon signed-rank
≥ 3 independent	One-way / Welch ANOVA	Kruskal–Wallis
≥ 3 matched	RM ANOVA / mixed model	Friedman
Post hoc	Tukey, Dunnett, Games–Howell	Dunn (Holm)
Association	Pearson r	Spearman ρ

Small n warning

With 3 vs 3 the smallest two-sided Mann–Whitney p is 0.10. Normality tests have little power at n ≤ 5; use QQ plots and what you know about the assay.

Statistics using R


```R
1 library(tidyverse); library(rstatix)
  d <- read_csv("data/stats_example.csv")     # group, value

2 d %>% group_by(group) %>% shapiro_test(value)  # normal?
  d %>% levene_test(value ~ group)               # equal SD?

3 d %>% t_test(value ~ group, detailed = TRUE)  # Welch
  d %>% t_test(value ~ group, var.equal = TRUE) # Student

4 d %>% wilcox_test(value ~ group)               # Mann-Whitney U

5 d %>% mutate(lv = log10(value)) %>% t_test(lv ~ group)
```

1 Load

One row per well: group and value.

2 Check the assumptions

Shapiro–Wilk per group, Levene across groups, plus QQ plots.

3 Parametric

Welch t-test by default; Student only when the SDs are equal.

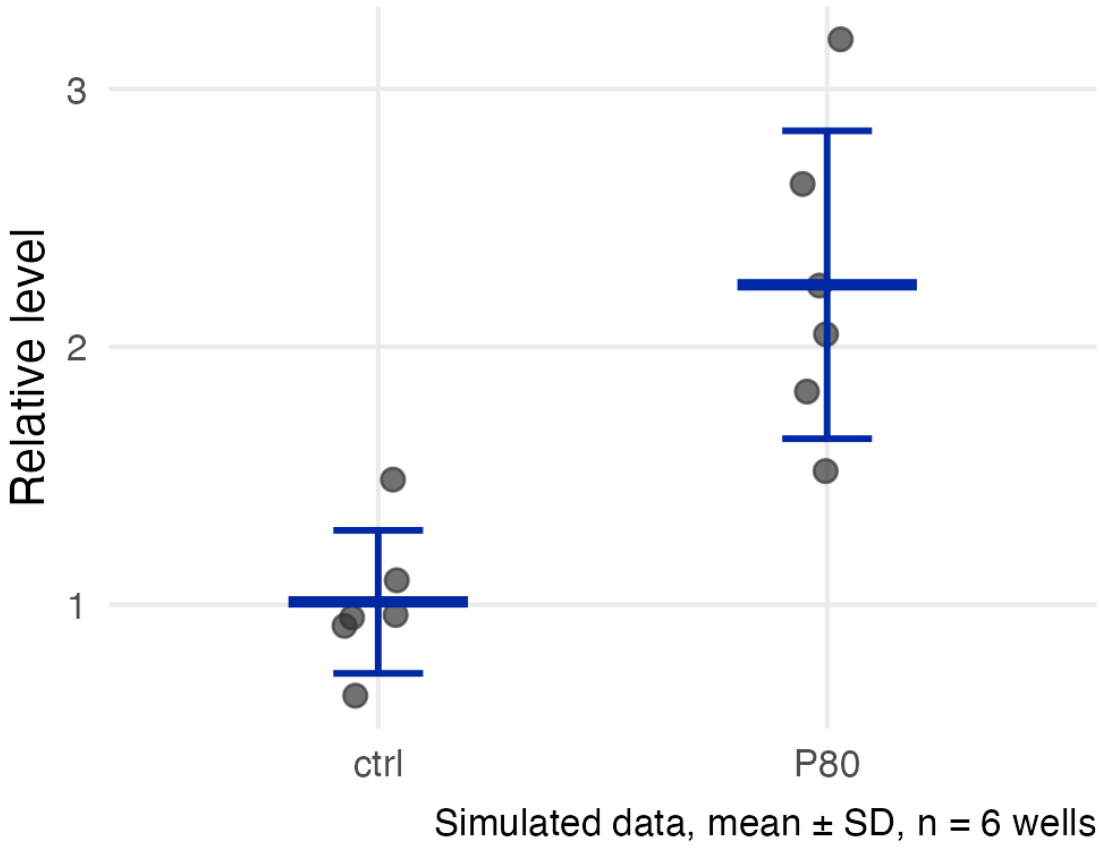
4 Nonparametric

Mann–Whitney U on the ranks of all 12 wells.

5 Skewed read-out

Log-transform, then the t-test again.

Result: both families agree



Test	Result
Shapiro–Wilk (ctrl / P80)	p = 0.49 / 0.91
Levene	p = 0.15
Welch t-test	t(7.1) = −4.58, p = 0.002
Mann–Whitney	U = 0, p = 0.002

P80 vs ctrl: higher

Difference 1.23 (95 % CI 0.60–1.86), log scale: p < 0.001

Ranks need replicates

U = 0: every P80 well above every control well. With 3 vs 3 the same pattern gives only p = 0.10.

MTT viability assay

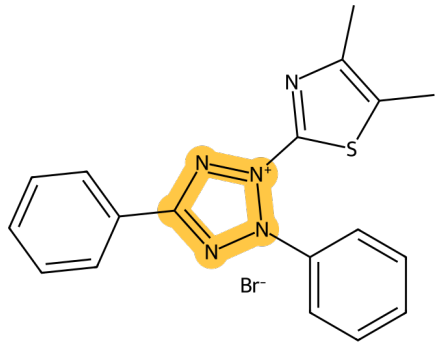
Theory, data structure, algorithm, implementation

02

MTT reports metabolic activity, not cell number

MTT

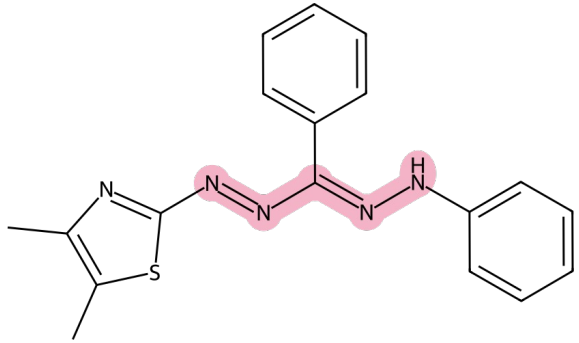
yellow, water-soluble tetrazolium salt



NAD(P)H oxidoreductases of living cells

Formazan

purple, insoluble crystals



dissolve in DMSO

Absorbance

540 nm, formazan dissolved in DMSO



more living cells → more formazan → darker purple → higher A540

Beer–Lambert

$A = \varepsilon \times c \times l$: absorbance is proportional to formazan, so to active cells (within the linear range of cell number).

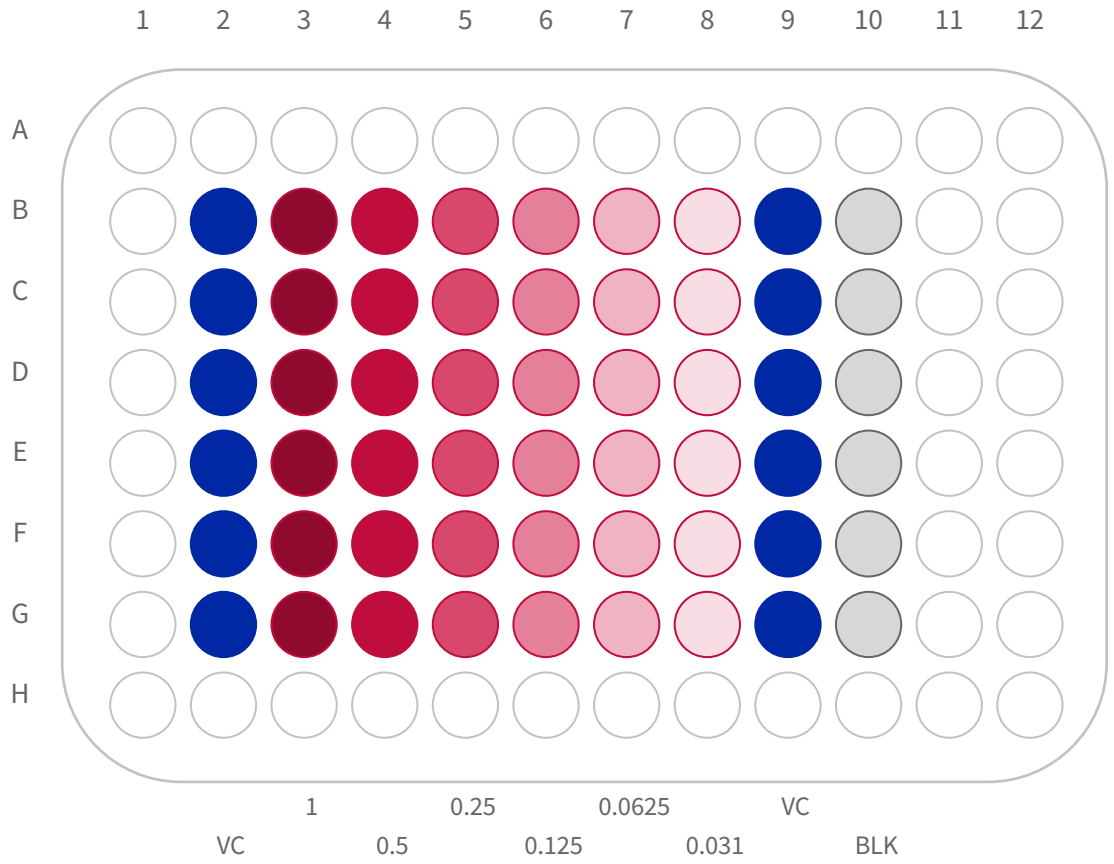
Activity, not number

Fewer cells and slower metabolism look the same. Confirm with a second read-out: LDH release, live/dead staining, cell count.

Interference

Reducing compounds (polyphenols, antioxidants) make formazan without cells: add a cell-free compound + MTT control.

Plate map becomes a long table



Cols 2 + 9 vehicle control (VC); **cols 3–8** P80 %; **col 10** blank
Rows B–G 6 replicate wells; 10,000 cells/well

well	condition	p80_pct	od540
B2	ctrl	0	0.728
B3	P80-1	1	0.095
B8	P80-0.03125	0.03125	0.702
B9	ctrl	0	0.683
B10	blank		0.094

data/mtt_raw.csv: 54 rows = 9 columns × 6 wells, one plate

One row per well

Well, condition and concentration travel with every reading; nothing depends on positions.

Unit = well

One plate, one experiment → n = 6 wells per concentration, 12 controls.

From optical density to a dose-response curve

$$\text{Viability (\%)} = (OD_{well} - \text{mean } OD_{blank}) / (\text{mean } OD_{control} - \text{mean } OD_{blank}) \times 100$$

1. **Blank**: mean of the medium + MTT + DMSO wells (no cells)
2. **Normalise**: (OD540 − blank) ÷ mean of the 12 control wells, × 100
3. **Check the plate**: control column 2 vs 9 should agree
4. **Assumptions**: residual QQ plot / Shapiro–Wilk; Levene
5. **Test**: ANOVA + Dunnett vs control; unequal SDs: Welch + Games–Howell
6. **Curve**: 4-parameter log-logistic fit → IC50 with 95 % CI

Why Dunnett, not 6 t-tests

Six comparisons with the same control: six t-tests at 0.05 inflate false positives; Dunnett keeps the family-wise error at 5 %.

Dead wells have almost no spread

Dunnett pools one SD over all groups. When SDs differ: Welch ANOVA + Games–Howell.

MTT analysis using R


```R
mtt   <- read_csv("data/mtt_raw.csv")
blank <- mean(mtt$od540[mtt$condition == "blank"])

wells <- mtt %>% filter(condition != "blank") %>%
  mutate(od = od540 - blank,
         viability = 100 * od / mean(od[p80_pct == 0]),
         dose = factor(p80_pct))          # "0" = control

wells %>% filter(p80_pct == 0) %>%                # col 2 vs 9
  t_test(viability ~ column)
wells %>% levene_test(viability ~ dose)           # equal SD?

fit <- aov(viability ~ dose, data = wells)
emmeans(fit, trt.vs.ctrl ~ dose)                 # Dunnett
wells %>% games_howell_test(viability ~ dose)     # unequal SD

curve <- drc::drm(viability ~ p80_pct, data = wells,
                  fct = drc::LL.4())
drc::ED(curve, 50, type = "absolute", interval = "delta")
```

1. **Blank**
   Medium + MTT + DMSO without cells.

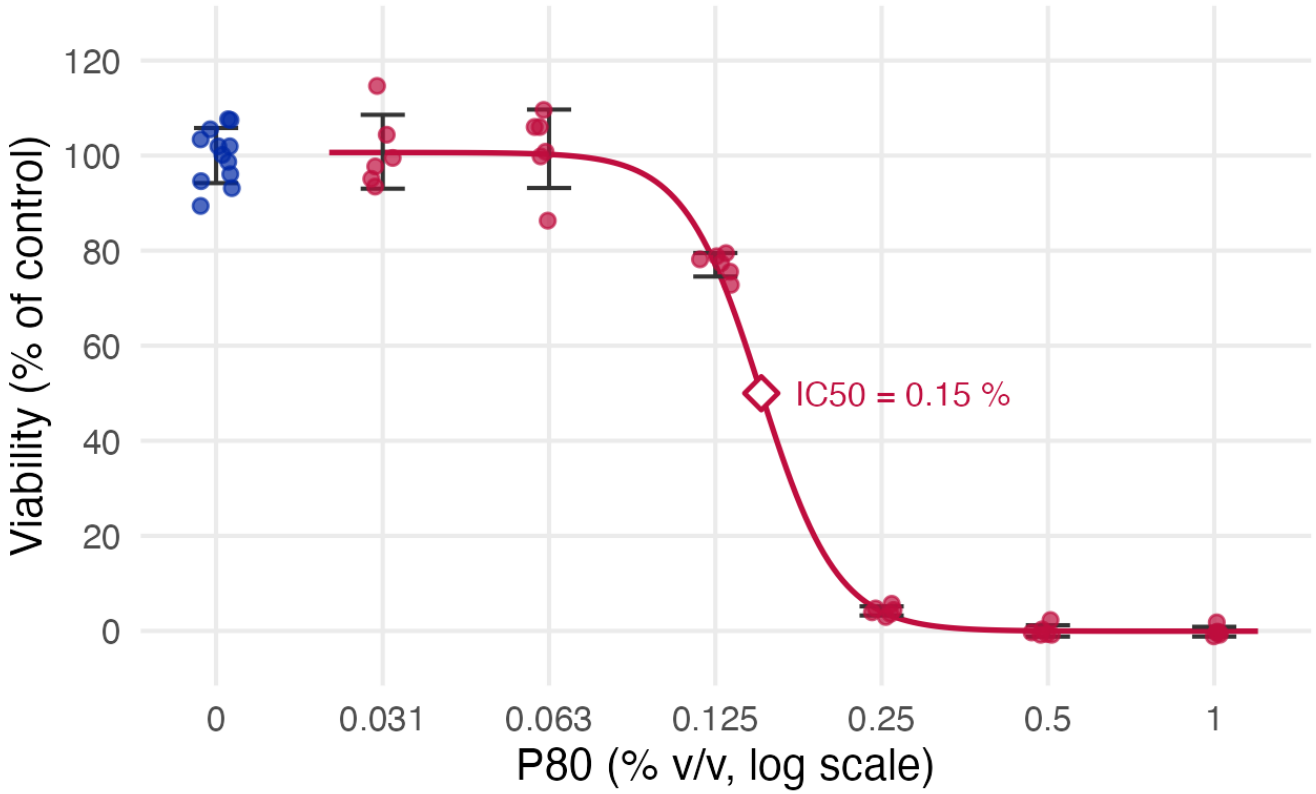
2. **% of control**
   Each well relative to the control mean.

3. **Check plate and spread**
   Columns 2 vs 9: p = 0.75. SD 6–8 % alive, 1 % dead: Levene p = 0.01.

4. **Each dose vs control**
   Both: ≥ 0.125 % lower ($p < 0.001$); 0.03125 and 0.0625 % not.

5. **IC50**
   4-parameter log-logistic fit; 50 % viability.

Result: MTT dose response



Simulated data, points = wells (n = 6; control 12); mean ± SD; line = 4-parameter fit

Mean ± SD, n = 6 wells: 0.031 % 101 ± 8; 0.0625 % 101 ± 8; 0.125 % 77 ± 2; 0.25 % 4 ± 1; 0.5 % 0 ± 1; 1 % 0 ± 1

0.15 %

IC50 (95 % CI 0.14–0.16 %), steep 4-parameter fit

≥ 0.125 %

lower than control (Games–Howell); from 0.25 % almost no living cells

100, 94, 0 %

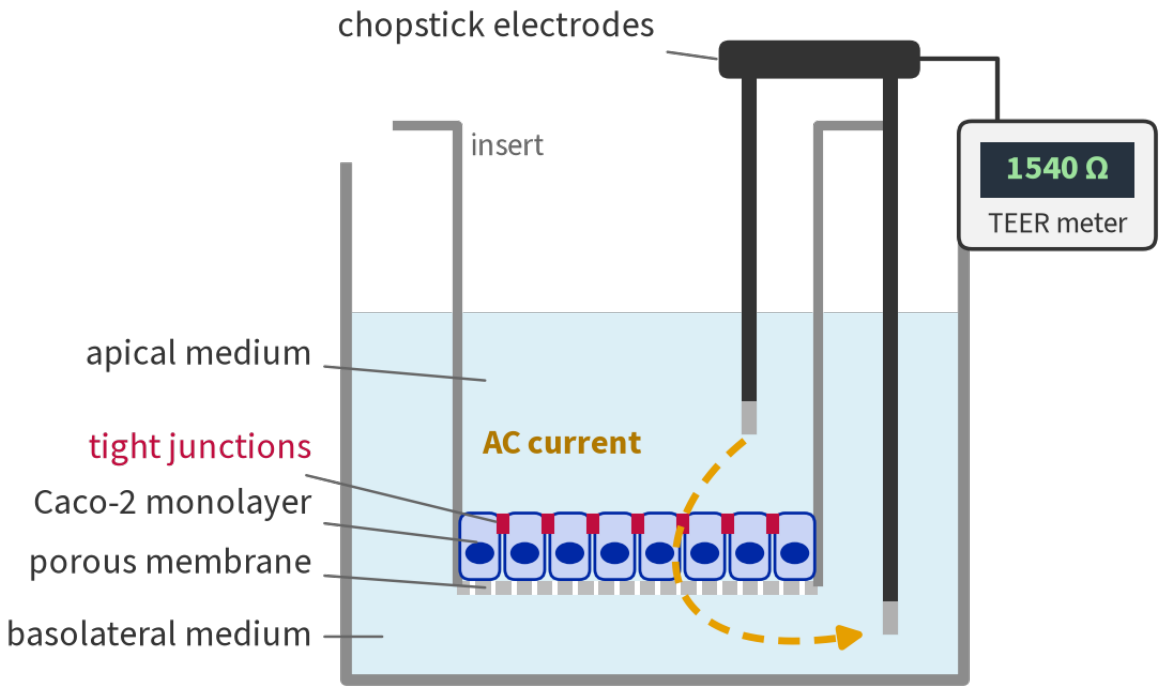
predicted viability at the barrier doses 0.02, 0.1, 0.5 %

Transepithelial electrical resistance (TEER)

Theory, data structure, algorithm, implementation

03

TEER is Ohm’s law across a cell layer



$R_{measured} = R_{blank}$ (medium + filter) + R_{cells}

R_{cells}: paracellular (tight junctions) in parallel with transcellular

Ohm’s law

R = V / I. A small AC current (≈ 12.5 Hz square wave) avoids electrode polarisation and harm to the cells.

Junctional ion flow

Mainly ion flow through tight junctions; it falls when junctions open.

Area-corrected: Ω cm²

Resistance is inversely proportional to area: multiplying by the membrane area makes inserts comparable.

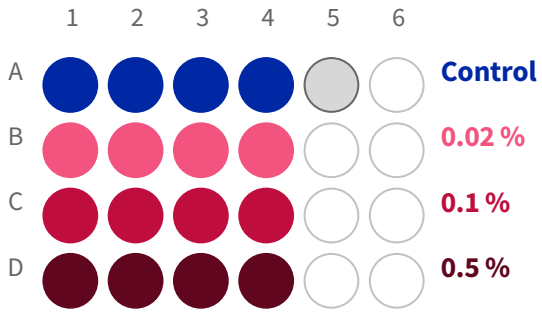
Sensitive to

Temperature (R falls as medium warms), electrode position and depth, medium changes.

Readings × inserts × time points

insert	condition	time_h	reading	r_ohm
A1	ctrl	0	2	1533
A1	ctrl	0	3	1539
A1	ctrl	24	2	1554
A1	ctrl	24	3	1547
A5	blank	0	2	180
A5	blank	0	3	184
…	…	…	…	…

data/teer_raw.csv: 102 rows = 17 inserts × 3 times × 2 readings



A5 = empty insert (blank), 24-well Transwell plate

Membrane area: Corning Transwell

Plate	Insert Ø	Area
24-well	6.5 mm	**0.33 cm²**
12-well	12 mm	1.12 cm²
6-well	24 mm	4.67 cm²

Technical level

2nd and 3rd reading per insert and time → average.

Blank

One empty insert (A5) with medium, read at every time point.

Repeated measures

Each insert is read at 0, 24 and 48 h: time points are not independent.

Independent inserts

4 conditions × 4 inserts on one plate: n = 4.

From ohms to Ω cm² and % of baseline

$$\text{TEER}\ (\Omega\ \text{cm}^2) = (R_{\text{insert}} - R_{\text{blank}}) \times A \qquad A = 0.33\ \text{cm}^2\ \text{(24-well insert)}$$

1. **Average readings**: mean of the 2nd and 3rd reading
2. **Subtract blank**: empty insert, same time point
3. **× area**: Ω → Ω cm² (0.33 cm²)
4. **% of baseline**: divide by the insert's own 0 h value
5. **End-point** at 48 h: ANOVA + Dunnett vs control (unequal SDs: Games–Howell)
6. **Time course**: mixed model, condition × time, insert as random effect

Worked example (SIAF data, Ogulur et al., Allergy 2023)

(1548 − 175 Ω) × 0.33 cm² = 1373 Ω × 0.33 cm² = 453 Ω cm²

Each insert is its own baseline

Inserts start at different TEER (here 370–550 Ω cm²): each insert becomes its own reference before P80 is added.

TEER analysis using R

R

```r
# 1
teer <- read_csv("data/teer_raw.csv") %>%
  group_by(insert, condition, time_h) %>%
  summarise(r_ohm = mean(r_ohm), .groups = "drop")

# 2
blank <- teer %>% filter(condition == "blank") %>%
  select(time_h, r_blank = r_ohm)
teer <- teer %>% filter(condition != "blank") %>%
  left_join(blank, by = "time_h") %>%
  mutate(teer = (r_ohm - r_blank) * 0.33)          # ohm * cm2

# 3
teer <- teer %>% group_by(insert) %>%
  mutate(pct = 100 * teer / teer[time_h == 0])

# 4
end48 <- teer %>% filter(time_h == 48)
end48 %>% levene_test(pct ~ condition)          # p = 0.002
end48 %>% games_howell_test(pct ~ condition)    # vs ctrl

# 5
fit <- lmer(teer ~ condition * factor(time_h) + (1 | insert),
            data = teer)
```

1 Average the readings

2nd and 3rd reading → one value per insert and time.

2 Blank, × area

Empty insert at the same time; × 0.33 cm^2 → Ω cm^2.

3 % of baseline

Each insert is its own reference (0 h).

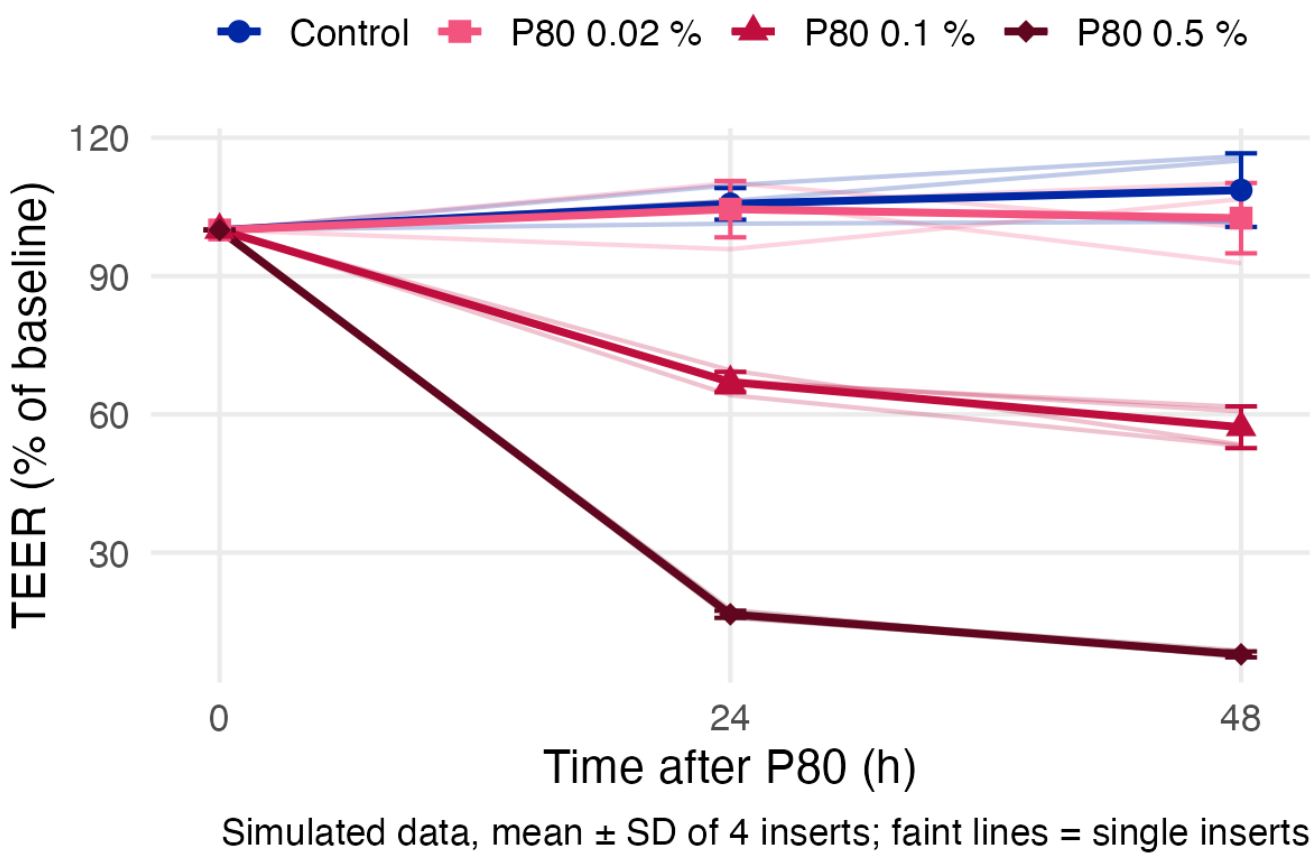
4 End-point at 48 h

Each P80 concentration vs control; SDs differ → Games–Howell.

5 Whole time course

Mixed model: inserts are read repeatedly.

Result: TEER over 48 h



Simulated data, mean ± SD of 4 inserts; faint lines = single inserts

57 %

0.1 % at 48 h (control 109 %), Games–Howell $p < 0.001$; 0.02 %: 103 %, $p = 0.70$

8 %

0.5 % at 48 h; the MTT shows no living cells at 0.5 %

F = 83

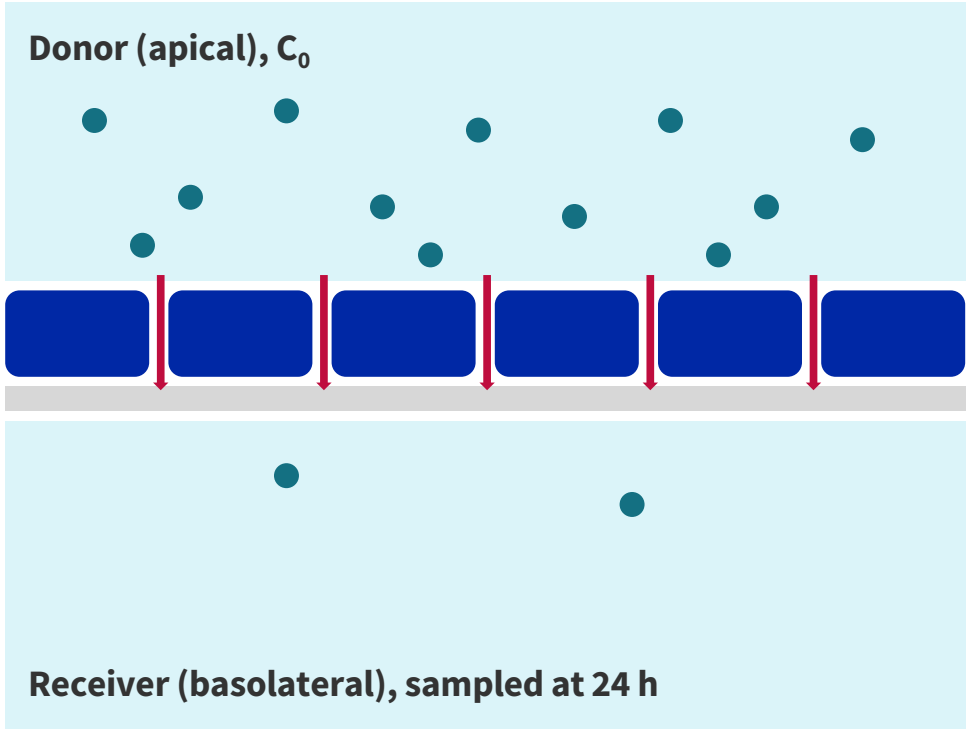
condition × time, mixed model, $F(6, 24)$; $p < 0.001$

Paracellular flux (FITC-dextran 4 kDa)

Theory, data structure, algorithm, implementation

04

Flux measures tracer transport across the barrier



Gaps between cells = paracellular route (red arrows).

After 24 h both sides are sampled: 100 µL in duplicate wells.

Fick's first law

$J = (dQ/dt) / A = P_{app} \times C_0$. One end-point: $dQ/dt \approx Q / t$ while the receiver stays below ~10 % of the donor.

Apparent permeability

P_{app} in cm/s normalises the rate for area and donor concentration, comparable across inserts and labs.

Tracer size selects the pathway

Na-fluorescein (376 Da), Lucifer yellow (457 Da): small solutes. FITC-dextran 4 kDa (FD4) and larger: leak pathway.

Read together with TEER

TEER ↓ with unchanged FD4 flux → pore (ion) pathway; FD4 flux ↑ → leak pathway or lesions.

Two tables and a set of constants

Standards (duplicates)

conc_ug_ml	rfu
0	52
0	47
0.39	409
0.39	427
0.78	786
…	…
12.5	12 238
25	23 401
25	23 821

Samples (64 rows = 16 inserts × 2 sides × 2)

insert	side	dilution	rfu
A1	basolateral	1	1087
A1	basolateral	1	1103
A1	apical	50	9852
A1	apical	50	9716
…	…	…	…

Stay inside the standards

- Range 0.39–25 µg/mL: apical samples (~500 µg/mL) diluted 1 : 50
- Duplicate wells → average; they are not n
- Units in column names: conc_ug_ml, time in s

Constants

- $A = 0.33\ cm^2$
- $C_0 = 500\ µg/mL$ (0.5 mg/mL)
- $V_{api} = 0.3\ mL$
- $V_{baso} = 0.6\ mL$
- $t = 24\ h = 86\ 400\ s$

Same plate, same medium

Standards in cDMEM next to the samples; blank = medium only.

From fluorescence to apparent permeability

$$P_{app} = (C_{baso} \times V_{baso}) / (t \times A \times C_0) \qquad \mu g \div (s \times cm^2 \times \mu g/cm^3) = cm/s$$

1. **Standard curve**: linear fit RFU = a + b × C over 0.39–25 µg/mL; R^2
2. **Concentration**: mean of duplicates; C = (RFU − a) / b × dilution
3. **Amount crossed**: $C_{baso} \times V_{baso}$; % of the dose ($C_0 \times V_{api}$)
4. **Recovery**: apical + basolateral ≈ 100 % of the dose
5. **P_{app}**: end-point, in cm/s
6. **Test**: $\log_{10} P_{app}$: ANOVA + Dunnett vs control; Spearman with TEER

498 / 517 nm are settings, not two readings

Excitation 498 nm and emission 517 nm define one fluorescence reading; concentration comes from the standard curve, not from a subtraction.

Sink conditions

Keep the transported fraction below ~10 % of the dose so the gradient stays constant (here ≤ 9 %: close to the limit at 0.5 % P80).

Flux analysis using R


```r
cal <- lm(rfu ~ conc_ug_ml, data = std)            # R2 = 0.9996

conc <- smp %>% group_by(insert, condition, side, dilution) %>%
  summarise(rfu = mean(rfu), .groups = "drop") %>%  # duplicates
  mutate(conc = (rfu - coef(cal)[1]) / coef(cal)[2] * dilution)

papp <- conc %>% select(insert, condition, side, conc) %>%
  pivot_wider(names_from = side, values_from = conc) %>%
  mutate(crossed = basolateral * 0.6,              # ug
         papp = crossed / (86400 * 0.33 * 500))    # cm/s

papp <- papp %>% mutate(log_papp = log10(papp))
fit  <- aov(log_papp ~ condition, data = papp)
emmeans(fit, trt.vs.ctrl ~ condition)              # Dunnett

both %>% cor_test(teer, papp, method = "spearman")  # TEER 48 h
```

1. **Standard curve**
   7 standards, 25 → 0.39 µg/mL.

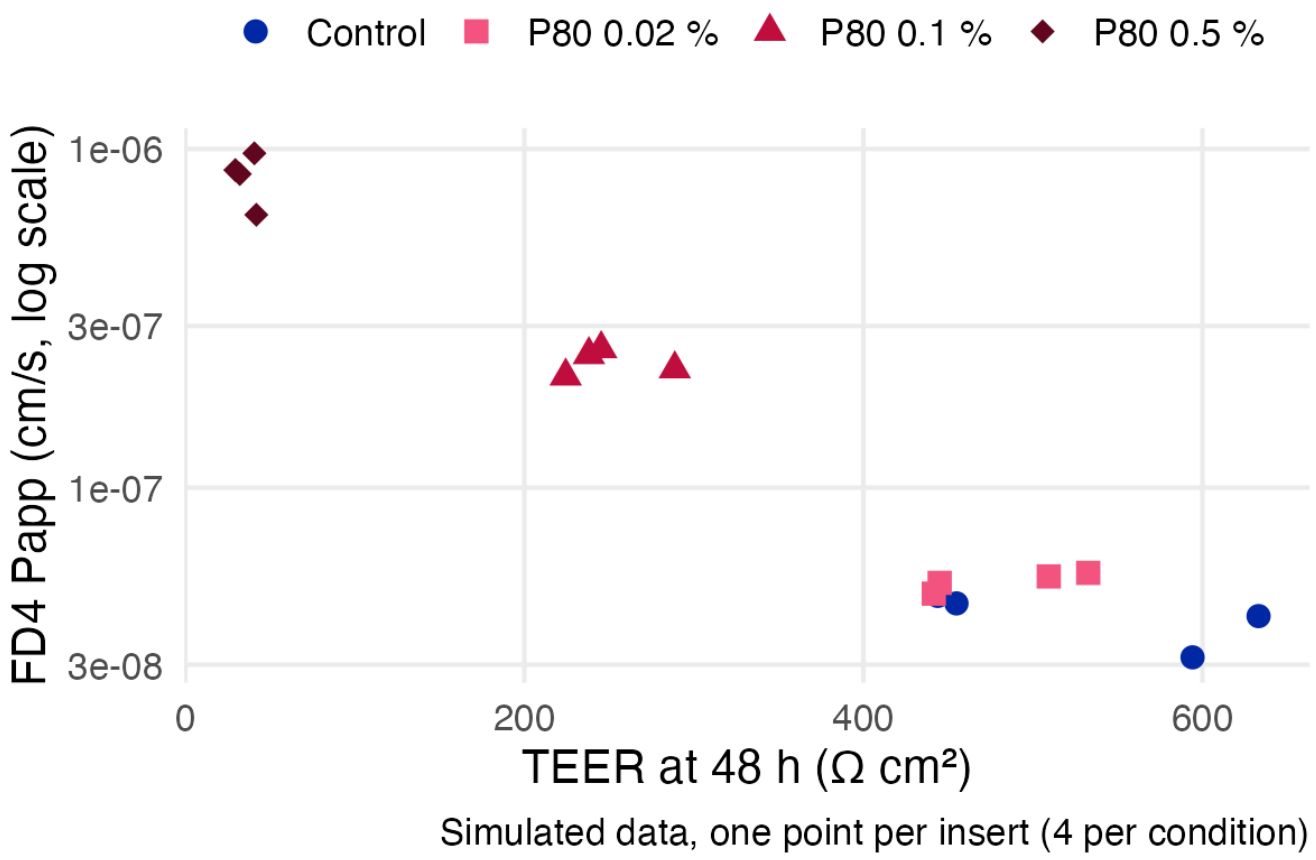
2. **RFU → µg/mL**
   Mean of duplicates; × dilution (apical 1:50).

3. **Amount → P_{app}**
   $C_{baso} \times V_{baso} / (t \times A \times C_0)$.

4. **Test on log scale**
   P_{app} spans orders of magnitude; $10^{estimate}$ = fold change.

5. **Relate to TEER**
   Spearman ρ across the 16 inserts.

Result: permeability and TEER



1.3, 5.7, 20 ×

P_{app} at 0.02, 0.1, 0.5 % vs control (geometric means)

p < 0.001

0.1 and 0.5 %, Dunnett on $\log_{10} P_{app}$; 0.02 %: p = 0.07

ρ = −0.89

TEER 48 h vs P_{app}, 16 inserts, p < 0.001

Before you report barrier data: a checklist

1. n = wells or inserts of one experiment (say so in every legend)
2. MTT: blank subtracted, both control columns agree, control = 100 %, IC50
3. ≥ 3 groups: ANOVA + Dunnett vs control; Games–Howell when SDs differ
4. Exact p-values, test named, effect with 95 % CI
5. TEER in Ω cm^2: blank subtracted, × 0.33 cm^2, % of baseline
6. Time course with a mixed model: inserts are read repeatedly
7. FD4: standards bracket every sample; dilutions and recovery stated
8. P_{app} with tracer, time, area and C_0; read together with TEER

Run everything yourself

```
BME364-G1-share/
├── README.md          start here
├── R/                 mac_linux/, windows/ (.Rmd)
├── data/              example data (.csv)
├── example_images/    4 real IF fields (.tif)
├── fiji/              measure_zo1.ijm
├── figures/           plots, written by R
├── results/           tables, written by R
├── manuals/           lab manuals (.docx)
└── slides/            both decks (.pptx)
```

R

```r
# R: R/mac_linux/ or R/windows/
# once: Source 00_install_packages.R
# knit in RStudio, in this order:
#   01_statistics.Rmd, 02_mtt.Rmd,
#   03_teer.Rmd, 04_flux.Rmd (needs 03)
```

Your own data from Davos

- R packages: rmarkdown, tidyverse, rstatix, lme4, lmerTest, emmeans, drc
- Replace the files in data/ with your own exports and keep the column names; the .Rmd files then reproduce every step of this workshop.